

Spark SQL

- Spark SQL is a Spark module for structured data processing.
 - Spark SQL can also be used to read data from an existing Hive installation
 - **When running SQL from within another programming language the results will be returned as a [Dataset/DataFrame](#)**
 - Execute Spark SQL command in command line or over ODBC/JDBC
 - Speed of spark SQL is comparable to that of Impala

Three APIs @ K-State VM1

- spark-sql
 - Create a table
 - Load data into a table
 - Use SQL commands

spark-sql demo

- Prereq
 - At vm prompt type: spark-sql
 - the following prompt means it is ready: spark-sql>
 - Each SQL line can form a separate line after >
 - Enter ; after > or at the end of a statement to execute
 - The SQL statements are NOT case sensitive
 - Highlight the SQL statements in an editor, CTRL + c to copy; at spark-sql> right-click to paste
 - CTRL + c to exit the shell
 - When spark-sql fails to run, at vm>jps to identify all java process, vm>kill xxxxx for the job number after the SparkSubmit job.

Flight data in 2015 is used for demonstration

- A csv file with three columns available in Canvas (2015-summary.csv)
- Upload the data file: 2015-summary.csv under VM1 /home/changs/.
- Rename the file to flights-2015-summary.csv for Databricks upload
- Note the third column name has been changed from “count” to “fnum”

Spark-sql commands (1)

- # first check if a table is in the default db

SHOW TABLES;

- # Create the table and load the data if the flights table does not exist

```
CREATE TABLE flights (  
  DEST_COUNTRY_NAME STRING,  
  ORIGIN_COUNTRY_NAME STRING,  
  fnum LONG)  
USING csv OPTIONS (header 'true', path 'file:///home/changs/2015-summary.csv');
```

What is the table name?

- # provide table information; note that the database is default

```
DESCRIBE TABLE flights;
```

- # Show all data

```
SELECT * FROM flights;
```

Database and tables

- A database contains many tables
- In spark-sql, the default db is called default.

#Database commands

```
CREATE DATABASE some_db;  
USE some_db;
```

#Show what tables are in a database

```
SHOW TABLES;
```

#Show the current db

```
SELECT current_database();
```

#Drop or delete a table

```
DROP TABLE flights;
```

Wildcard in WHERE

```
SELECT ORIGIN_COUNTRY_NAME  
FROM flights  
WHERE ORIGIN_COUNTRY_NAME like 'A%';
```

- What is the answer that above SELECT statement gives?

#ORDER BY examples

```
SELECT ORIGIN_COUNTRY_NAME, DEST_COUNTRY_NAME, fnum  
FROM flights  
WHERE fnum > 1000  
ORDER BY ORIGIN_COUNTRY_NAME;
```

```
SELECT ORIGIN_COUNTRY_NAME, DEST_COUNTRY_NAME, fnum  
FROM flights  
WHERE fnum > 1000  
ORDER BY ORIGIN_COUNTRY_NAME, DEST_COUNTRY_NAME;
```

```
SELECT ORIGIN_COUNTRY_NAME, DEST_COUNTRY_NAME, fnum  
FROM flights  
WHERE fnum > 1000  
ORDER BY ORIGIN_COUNTRY_NAME, fnum DESC;
```

#subqueries

#Top five destination countries

```
select dest_country_name from flights  
group by dest_country_name  
order by sum(fnum) desc limit 2;
```

subqueries demo

```
SELECT * from flights
```

```
WHERE origin_country_name IN (select dest_country_name from  
flights group by dest_country_name order by sum(fnum) desc limit 2);
```

#Aggregation example: sum, avg, count

```
SELECT avg(fnum)
```

```
FROM flights
```

```
WHERE DEST_COUNTRY_NAME='United States';
```

- What is the answer that above SELECT statement gives?

```
SELECT DEST_COUNTRY_NAME AS DC, SUM(fnum) AS Total_In_Flights
```

```
FROM flights
```

```
GROUP BY DC
```

```
ORDER BY Total_In_Flights;
```

- What is the answer that above SELECT statement gives?

Reference: SQL SELECT Syntax

```
SELECT [ALL|DISTINCT] named_expression[, named_expression, ...]  
  FROM relation[, relation, ...]  
  [lateral_view[, lateral_view, ...]]  
  [WHERE boolean_expression]  
  [aggregation [HAVING boolean_expression]]  
  [ORDER BY sort_expressions]  
  [CLUSTER BY expressions]  
  [DISTRIBUTE BY expressions]  
  [SORT BY sort_expressions]  
  [WINDOW named_window[, WINDOW named_window, ...]]  
  [LIMIT num_rows]
```

- Chambers chapter 10 pp 190
- **Select syntax:** <https://docs.databricks.com/spark/latest/spark-sql/language-manual/select.html>